Rational rotations of three dimensional Minkowski space: rotations, boosts and spinors

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Abstract

We give an algebraic and projective account of the rotations and boosts of three dimensional Minkowski space over a field of characteristic not two. No angles, rapidities, circular or hyperbolic functions, or unit-vector normalizations are required, and we bypass also Lie theory. On an affine chart the main objects are the **half spread**, which measures the amount of a transformation, and the **anchor**, a metrically antisymmetric matrix which determines every rotation except a half turn. The associated **cap** is a Cayley transform. Passing from anchors to projective proportions removes the exceptional cases. The underlying four dimensional algebra is the split quaternion algebra, here called the **dihedron algebra**, and is canonically $M_2(\mathbb{F})$. Its non-null projective points form $PGL_2(\mathbb{F})$, which maps bijectively and multiplicatively onto $SO(1,2,\mathbb{F})$. Over $\mathbb{Q}$ the square class of the determinant is the spinor norm and is onto $\mathbb{Q}^*/(\mathbb{Q}^*)^2$, so rational determinant-one matrices reach only a subgroup of infinite index. Over $\mathbb{R}$ the positive-determinant component gives the usual $SL_2(\mathbb{R})$ double cover of the proper orthochronous Lorentz group. The null quadric is the rank-one locus in M_2 and its two rulings are the two projective spinor families. The theory is the Minkowski companion to the Euclidean projective theory of [11].

1 Introduction

Three dimensional Minkowski space [6] has coordinates (t,r,s) and quadrance

$$Q(t,r,s) = t^2 - r^2 - s^2.$$

Its determinant-one linear isometries form $SO(1,2)$. Over the real numbers this group contains the ordinary spatial rotations, the Lorentz boosts and the parabolic or null rotations.

The standard real theory uses angles for spatial rotations and rapidities for boosts and considers this a chapter in Lie theory. This provides a useful analytic orientation, but these concepts are not needed for the algebra. The formulas of Euler, Rodrigues and Cayley [1], [2], [3] already suggest a description using only field arithmetic; Rodrigues' composition law in particular predates Hamilton's quaternions [9], [8]. The purpose of this paper is to make that description projective and to keep it over the ground field, in the spirit of rational trigonometry and universal hyperbolic geometry [10], [15]. It means that computations are generally exact.

The main affine objects are the same as in the Euclidean companion [11]. The **anchor** is a metrically antisymmetric matrix A_v attached to a vector v, defined in Section 3. The **cap** of an anchor A,

$$c(A) = (I + A)(I - A)^{-1},$$

is a Cayley transform. The **half spread** of a transformation ρ,

$$p(\rho) = \frac{3 - \operatorname{tr}\rho}{4},$$

measures the amount of the transformation. For a cap with anchor vector v,

$$p = \frac{Q(v)}{1 + Q(v)}.$$

Composition of transformations becomes a rational operation on anchor vectors. Projectively, a non-null proportion $(x : v) = (x : t : r : s)$, that is a non-zero four-vector up to scaling with $x^2 + Q(v) \neq 0$, determines the transformation

$$\rho(x : v) = I + \frac{2(xA_v + A_v^2)}{x^2 + Q(v)},$$

which is the cap of $A_{v/x}$ when $x \neq 0$.

Here is the simplest physical example. Take the projective proportion

$$(3 : 0 : 0 : 1).$$

It gives

$$\rho(3 : 0 : 0 : 1) = \frac{1}{4}\begin{pmatrix} 5 & 3 & 0 \\ 3 & 5 & 0 \\ 0 & 0 & 4 \end{pmatrix}.$$

This matrix preserves $t^2 - r^2 - s^2$, has determinant 1, and fixes the spacelike line $[(0, 0, 1)]$. It is the usual boost in the r direction with velocity $3/5$. Its half spread is

$$p = \frac{-1}{3^2 - 1} = -\frac{1}{8}.$$

The inverse is represented by $(3 : 0 : 0 : -1)$. Composition with another transformation is multiplication of two 2×2 matrices, or equivalently a four component product of proportions.

The classical dictionary

For comparison only, work over $\mathbb{R}$. If a rotation has timelike unit axis n and classical angle θ, then on the affine chart

$$v = n \tan\frac{\theta}{2}, \qquad p = \sin^2\frac{\theta}{2}.$$

These are the same formulas as in the Euclidean companion.

For a standard boost let e be a spacelike unit vector fixed by the transformation, so $Q(e) = -1$, and write the anchor vector as ae. If φ is the classical rapidity and β the physical velocity, then

$$a = \tanh \frac{\varphi}{2}, \qquad \beta = \tanh \varphi = \frac{2a}{1+a^2},$$

while

$$p = -\frac{a^2}{1-a^2} = -\sinh^2 \frac{\varphi}{2}.$$

These identities connect the algebraic parameters with the usual physical ones. They are not definitions here.

The distinction between the anchor parameter a and the physical velocity β is important. Collinear anchors compose by

$$a_{12} = \frac{a_1 + a_2}{1 + a_1 a_2},$$

and substituting $\beta = 2a/(1+a^2)$ gives the Einstein law [5]

$$\beta_{12} = \frac{\beta_1 + \beta_2}{1 + \beta_1 \beta_2}.$$

Thus velocity addition follows from the projective composition law.

The Lorentz group and its spin cover

The four projective coordinates have a particularly simple matrix meaning:

$$\mu(x,t,r,s) = \begin{pmatrix} x+s & r-t \\ r+t & x-s \end{pmatrix}, \qquad \det \mu = x^2 + t^2 - r^2 - s^2.$$

This is the split quaternion algebra, also called the coquaternion algebra; we use the term **dihedron algebra** [12]. Projectivizing the invertible matrices gives

$$PGL_2(\mathbb{F}) \cong SO(1,2,\mathbb{F}).$$

Over $\mathbb{R}$ the sign of the determinant separates the two components of $SO(1,2,\mathbb{R})$. Positive determinant corresponds to the proper orthochronous component $SO^+(1,2)$, which preserves the direction of time (Section 11.1). After normalization to determinant one this gives the familiar two-to-one map

$$SL_2(\mathbb{R}) \longrightarrow SO^+(1,2).$$

Negative determinant gives the non-orthochronous component of $SO(1,2,\mathbb{R})$. This distinction will reappear algebraically through the half spread.

Over $\mathbb{Q}$ a rational projective matrix need not admit a rational determinant-one representative. The obstruction is the square class of its determinant, which is the spinor norm. This gives a sharper distinction between projectivization and normalization than in the real theory.

The null proportions satisfy

$$x^2 + t^2 - r^2 - s^2 = 0.$$

In the matrix model these are exactly the rank-one matrices. Their projective quadric has two rulings, corresponding to the column and row spinor lines. This makes spinors part of the projective geometry rather than an additional analytic construction.

The paper follows the same general order as the Euclidean companion. Sections 2 and 3 develop the vector and anchor algebra. Section 4 treats reflections and half turns. Sections 5–7 introduce the cap, half spread and anchor of a rotation. Section 8 gives an algebraic orientation test, and Section 9 gives the anchor product and velocity addition. Section 10 isolates the limits of the affine chart. Section 11 passes to the matrix algebra and proves the projective rotation theorem. Section 12 studies the spinor norm, causal type, spin and the rational obstruction.

2 Quadrances, spreads and vector algebra

Throughout, $\mathbb{F}$ is a field of characteristic not two, with $\mathbb{F} = \mathbb{Q}$ primary. Vectors are row vectors $v \equiv (t, r, s) \in \mathbb{F}^3$, and the metrical structure is the **Gram matrix**

$$G \equiv \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

Note $G^2 = I$ and $\det G = 1$; the second fact, shared with the Euclidean signature $(3, 0)$, underlies the parallel with [11]. For a non-zero vector v we write $[v]$ for the **line** it spans.

Definition 1 *The **dot product** of v_1 and v_2 is*

$$\langle v_1, v_2 \rangle \equiv v_1 G v_2^T = t_1 t_2 - r_1 r_2 - s_1 s_2,$$

*and the **quadrance** of $v = (t, r, s)$ is $Q(v) \equiv \langle v, v \rangle = t^2 - r^2 - s^2$. Vectors are **perpendicular** when $\langle v_1, v_2 \rangle = 0$, and v is **null** when $Q(v) = 0$. A non-null vector is **timelike** when $Q(v)$ is positive and **spacelike** when it is negative, over an ordered field.*

Definition 2 *The **adjoint** of a row vector is $v^\dagger \equiv G v^T$, and the **adjoint** of a 3×3 matrix M is $M^\dagger \equiv G M^T G$, so that $\langle v_1, v_2 \rangle = v_1 v_2^\dagger$ and $(MN)^\dagger = N^\dagger M^\dagger$.*

The adjoint replaces the transpose: passing to the Euclidean companion replaces every adjoint by a transpose, and the Euclidean theory is the case $G = I$.

Theorem 3 (Null vectors) *Over every field of characteristic not two there exist non-zero null vectors, for instance $(1, 1, 0)$. Over $\mathbb{Q}$ the null lines are exactly the lines of the Pythagorean cone $t^2 = r^2 + s^2$, parametrized by the projective line: every rational null line is $[(m^2 + n^2, m^2 - n^2, 2mn)]$ for a unique point $(m : n)$ of $P^1(\mathbb{Q})$.*

Proof. $Q(1, 1, 0) = 1 - 1 = 0$. A null vector with $t = 0$ has $r^2 + s^2 = 0$, hence over $\mathbb{Q}$ is zero; a non-zero rational null vector may be scaled to $t = m^2 + n^2$, and $(m^2 + n^2)^2 = (m^2 - n^2)^2 + (2mn)^2$ with the classical parametrization of Pythagorean triples gives the bijection. ∎

A non-zero null vector is perpendicular to itself. In the Euclidean theory over $\mathbb{Q}$ there are no null vectors; here they are rational and organize much of what follows.

Definition 4 *The **spread** between non-null vectors v_1, v_2 is*

$$s(v_1, v_2) \equiv 1 - \frac{\langle v_1, v_2 \rangle^2}{Q(v_1)\, Q(v_2)}.$$

The spread depends only on the lines spanned. For non-null vectors, $s(v_1, v_2) = 1$ precisely when v_1 and v_2 are perpendicular. Over $\mathbb{Q}$ the spread is not confined to $[0, 1]$; Section 12.1 relates the causal character of an axis to the range of half spreads of the rotations about it.

Definition 5 *The **cross product** of v_1 and v_2 is*

$$v_1 \times v_2 \equiv (r_2 s_1 - r_1 s_2,\ s_1 t_2 - t_1 s_2,\ t_1 r_2 - t_2 r_1).$$

Compared with the Euclidean cross product, the first component is negated. With this choice the cross product lemma $wA_v = v \times w$ of Section 3 holds for the Minkowski anchor, exactly as in the Euclidean theory.

Theorem 6 (Vector algebra) *For any vectors u, v, w, z in $\mathbb{F}^3$:*

1. *(Scalar triple product) $\langle u \times v, w \rangle = -\det \begin{pmatrix} u \\ v \\ w \end{pmatrix}$, and in particular $\langle u \times v, u \rangle = \langle u \times v, v \rangle = 0$;*
2. *(Vector triple product) $u \times (v \times w) = \langle u, w \rangle v - \langle u, v \rangle w$;*
3. *(Binet–Cauchy) $\langle u \times v, w \times z \rangle = \langle u, w \rangle \langle v, z \rangle - \langle u, z \rangle \langle v, w \rangle$.*

Proof. Each side is multilinear, so it suffices to check standard basis vectors, a finite computation. ∎

These identities, and the whole vector algebra of this section, extend to an arbitrary non-degenerate symmetric bilinear form on $\mathbb{F}^3$, with the corresponding generalised vector product; see the thesis of Notowidigdo [13] and [14]. The Euclidean form of [11] and the present Minkowski form are two instances.

The minus sign in part 1 is the basic signature change from the Euclidean vector algebra. The remaining formulas keep the same form.

Example 7 *Take $u = (1, 2, 3)$, $v = (2, -1, 1)$, $w = (1, 0, 0)$, so $u \times v = (-5, 5, -5)$. The scalar triple product is*

$$\langle u \times v, w \rangle = -5 = -\det \begin{pmatrix} u \\ v \\ w \end{pmatrix},$$

and $\langle u \times v, u \rangle = \langle u \times v, v \rangle = 0$. For the vector triple product, $v \times w = (0, 1, 1)$ and

$$u \times (v \times w) = (1, -1, 1) = \langle u, w \rangle v - \langle u, v \rangle w = (2, -1, 1) - (1, 0, 0),$$

since $\langle u, w \rangle = \langle u, v \rangle = 1$.

Theorem 8 (Lagrange identity) *For any v_1, v_2,*

$$\langle v_1, v_2\rangle^2 + Q(v_1 \times v_2) = Q(v_1)\, Q(v_2),$$

and consequently for non-null v_1, v_2,

$$s(v_1, v_2) = \frac{Q(v_1 \times v_2)}{Q(v_1)\, Q(v_2)}.$$

Proof. The Lagrange identity is Binet–Cauchy with $w = v_1$, $z = v_2$. ∎

Example 9 *For $v_1 = (1, 2, 3)$ and $v_2 = (2, -1, 1)$, we have $Q(v_1) = -12$, $Q(v_2) = 2$, $\langle v_1, v_2\rangle = 1$, $v_1 \times v_2 = (-5, 5, -5)$ of quadrance -25. The Lagrange identity reads*

$$1 - 25 = -24 = (-12)(2),$$

and $s(v_1, v_2) = 25/24 > 1$.

Two consequences recur. First, if v_1, v_2 are non-null but $v_1 \times v_2$ is null and non-zero, then $s(v_1, v_2) = 0$ though the lines are distinct: zero spread does not imply coincidence.

Example 10 *The vectors $v_1 = (0, 1, 0)$ and $v_2 = (1, 1, 1)$ have quadrances $-1, -1$ and dot product -1, so $s(v_1, v_2) = 1 - 1/1 = 0$; $v_1 \times v_2 = (-1, 0, -1)$ is null and non-zero. In the Euclidean theory this requires a finite field; here it is rational.*

Second, two perpendicular vectors can span a degenerate plane, and a cross product of independent vectors can vanish in quadrance: the null cone is woven through the vector algebra, and the non-nullity hypotheses of the coming sections are genuine rational restrictions.

Lemma 11 (Vanishing lemma) *Let g and h be polynomials in finitely many variables with rational coefficients, with h not the zero polynomial. If g vanishes at every rational point at which h does not vanish, then g is the zero polynomial.*

Proof. The product gh vanishes at every rational point. A polynomial over the infinite field $\mathbb{Q}$ which vanishes everywhere is the zero polynomial, by induction on the number of variables, since a non-zero polynomial in one variable has only finitely many roots. Hence $gh = 0$, and since the polynomial ring has no zero divisors and $h \neq 0$, $g = 0$. ∎

An identity with integer coefficients which holds on a non-empty rational generic locus therefore holds as a polynomial identity, and hence over every field.

3 Anchors

Definition 12 *The **anchor space** is $so(1, 2) \equiv \{A : A^\dagger = -A\}$, and the **anchor** of $v = (t, r, s)$ is*

$$A_v \equiv \begin{pmatrix} 0 & s & -r \\ s & 0 & t \\ -r & -t & 0 \end{pmatrix}.$$

The strictly upper triangular entries agree with the Euclidean anchor; the lower triangle is re-signed so that GA_v is antisymmetric, i.e. $A_v^\dagger = -A_v$. The map $v \mapsto A_v$ is a linear bijection from $\mathbb{F}^3$ onto $so(1,2)$, and the anchor acts on vectors as a cross product:

Lemma 13 (Cross product lemma) *For any v, w, $wA_v = v \times w$. In particular $vA_v = 0$.*

Proof. Both sides are bilinear, so it suffices to check the nine basis cases. ∎

Example 14 *For $v = (1,2,3)$,*

$$A_v = \begin{pmatrix} 0 & 3 & -2 \\ 3 & 0 & 1 \\ -2 & -1 & 0 \end{pmatrix}.$$

With $w = (2,-1,1)$,

$$wA_v = (-5,5,-5) = v \times w, \qquad vA_v = 0.$$

For row vectors v and w, $w^\dagger v$ denotes the 3×3 matrix with entries $\left(w^\dagger v\right)_{ij} = \left(Gw^T\right)_i v_j$.

Theorem 15 (Anchor product identity) *For any v, w,*

$$A_v A_w = w^\dagger v - \langle v, w \rangle I.$$

Proof. For any x, the cross product lemma and the vector triple product give

$$x\left(A_v A_w\right) = (v \times x) A_w = w \times (v \times x) = \langle w, x \rangle v - \langle w, v \rangle x = x\left(w^\dagger v - \langle v, w \rangle I\right)$$

using $xw^\dagger = \langle x, w \rangle$, and two matrices agreeing on all rows are equal. ∎

Corollary 16 *For any v, w:*

1. $A_v^2 = v^\dagger v - Q(v) I$;
2. $A_v A_w - A_w A_v = w^\dagger v - v^\dagger w = A_{w \times v}$;
3. $A_v A_w A_v = -\langle v, w \rangle A_v$;
4. $\operatorname{tr}\left(A_v A_w\right) = -2\langle v, w \rangle$.

Proof. For 1, set $w = v$. For 2, subtract the identity from its swap; the result is checked on basis vectors. For 3,

$$A_v A_w A_v = \left(w^\dagger v - \langle v, w \rangle I\right) A_v = w^\dagger (vA_v) - \langle v, w \rangle A_v = -\langle v, w \rangle A_v.$$

For 4, $\operatorname{tr}\left(w^\dagger v\right) = vGw^T = \langle v, w \rangle$, so the trace of the identity is $\langle v, w \rangle - 3\langle v, w \rangle$. ∎

These are the Euclidean statements with T replaced by $\dagger$.

Example 17 *For $v = (1,2,3)$ and $w = (2,-1,1)$, with $\langle v, w \rangle = 1$ and $w^\dagger = (2,1,-1)^T$,*

$$A_v A_w = \begin{pmatrix} 1 & 4 & 6 \\ 1 & 1 & 3 \\ -1 & -2 & -4 \end{pmatrix} = w^\dagger v - \langle v, w \rangle I.$$

Subtracting the reversed product gives $A_v A_w - A_w A_v = A_{w \times v}$ with $w \times v = (5,-5,5)$, and $\operatorname{tr}\left(A_v A_w\right) = -2 = -2\langle v, w \rangle$.

Definition 18 *The **anchor form** of anchors A, B is $A \cdot B \equiv -\frac{1}{2}\operatorname{tr}(AB)$, and the **quadrance** of an anchor is $Q(A) \equiv A \cdot A$.*

Theorem 19 (Isometry) *For any v, w, $A_v \cdot A_w = \langle v, w \rangle$ and $Q(A_v) = Q(v)$, so $v \mapsto A_v$ is an isometry between $\mathbb{F}^3$ with the relativistic dot product and $so(1,2)$ with the anchor form.*

Example 20 *Continuing with $v = (1,2,3)$, $w = (2,-1,1)$, $A_v \cdot A_w = -\frac{1}{2}(-2) = 1 = \langle v, w \rangle$, and since $\operatorname{tr}(A_v^2) = 24$, $Q(A_v) = -12 = Q(v)$.*

Theorem 21 (Characteristic polynomial of an anchor) *The characteristic polynomial of A_v is*

$$\det(XI - A_v) = X^3 + Q(v)X.$$

Proof. The X^2 coefficient is $-\operatorname{tr} A_v = 0$; the X coefficient is the sum of principal 2×2 minors, $(-s^2) + (-r^2) + t^2 = Q(v)$; and $\det A_v = \det G \cdot \det(GA_v) = 0$ since GA_v is antisymmetric of odd size. ■

Example 22 *For $v = (1,2,3)$, A_v has trace 0, sum of principal 2×2 minors -12, determinant 0, so $\det(XI - A_v) = X^3 - 12X = X^3 + Q(v)X$.*

The characteristic polynomial is identical to the Euclidean one, with the relativistic quadrance; every consequence drawn in [11] —invertibility of $I \mp A$, quadratic reduction of the cap, half spread identities—carries over verbatim.

4 Reflections and half turns

A **rotation** of Minkowski three space is a matrix $\rho \in M_3(\mathbb{F})$ satisfying

$$\rho G \rho^T = G, \qquad \det \rho = 1.$$

Thus $\rho^\dagger = \rho^{-1}$, and rotations preserve the dot product. The rotations form the group $SO(1,2,\mathbb{F})$, also written $SO(1,2)$; the matrices satisfying only $\rho G \rho^T = G$ form the group $O(1,2)$.

Lemma 23 (Fixed direction) *Every rotation $\rho \in SO(1,2,\mathbb{F})$ has eigenvalue 1.*

Proof. Using $\det \rho = 1$, then $\rho^{-1} = G\rho^T G$ with $\det G^2 = 1$, then transposition,

$$\det(\rho - I) = \det \rho \det\left(I - \rho^{-1}\right) = \det\left(G\left(I - \rho^T\right)G\right) = \det(I - \rho) = (-1)^3 \det(\rho - I).$$

The characteristic is not two, so $\det(\rho - I) = 0$. ■

A non-zero fixed vector determines a fixed line, which we call an **axis**. Its causal type may be timelike, spacelike or null.

Definition 24 *For a non-null vector v, the **reflection in the hyperplane of** v is*

$$\sigma_v \equiv I - \frac{2}{Q(v)} v^\dagger v.$$

Since $\sigma_{\lambda v} = \sigma_v$, the reflection depends only on the line $[v]$.

Theorem 25 (Reflections) *For a non-null vector v:*

1. *σ_v is self-adjoint, $\sigma_v^\dagger = \sigma_v$, and $\sigma_v^2 = I$;*
2. *$v\sigma_v = -v$;*
3. *$w\sigma_v = w$ precisely when $\langle w, v\rangle = 0$;*
4. *$\sigma_v \in O(1,2)$: $\langle w_1\sigma_v, w_2\sigma_v\rangle = \langle w_1, w_2\rangle$;*
5. *$\det \sigma_v = -1$.*

Proof. Self-adjointness is visible, and $\left(v^\dagger v\right)\left(v^\dagger v\right) = v^\dagger\left(vGv^T\right)v = Q(v)\, v^\dagger v$, so

$$\sigma_v^2 = I - \frac{4}{Q(v)} v^\dagger v + \frac{4}{Q(v)^2} Q(v)\, v^\dagger v = I,$$

giving 1, and 4 follows since $\sigma_v^\dagger \sigma_v = \sigma_v^2 = I$. For 2 and 3, $w\sigma_v = w - \frac{2\langle w,v\rangle}{Q(v)} v$. For 5: since v is non-null, v does not lie in the plane $v^\perp \equiv \{w : \langle w, v\rangle = 0\}$, so $\mathbb{F}^3 = [v] \oplus v^\perp$, and in a basis adapted to this decomposition the matrix of σ_v is $\operatorname{diag}(-1,1,1)$ by 2 and 3. ■

Non-nullity is now a rational restriction: for the null vector $v = (1,1,0)$, the plane $v^\perp = \{(t,t,s)\}$ contains v, and there is no reflection interchanging the sides of that plane along v. Null lines have no mirrors.

Example 26 *The vector $v = (2,1,1)$ is timelike with $Q(v) = 2$, and since $2/Q(v) = 1$ its reflection is*

$$\sigma_v = I - v^\dagger v = \begin{pmatrix} -3 & -2 & -2 \\ 2 & 2 & 1 \\ 2 & 1 & 2 \end{pmatrix}.$$

It reverses v: $v\sigma_v = (-2,-1,-1) = -v$; it fixes $w = (1,1,1)$, for which $\langle w, v\rangle = 0$; and $\det \sigma_v = -1$.

Definition 27 *For a non-null vector v, the **half turn** about $[v]$ is*

$$\rho_{[v]} \equiv -\sigma_v = -I + \frac{2}{Q(v)} v^\dagger v.$$

Theorem 28 (Half turns) *For a non-null vector v: $\rho_{[v]} \in SO(1,2)$, fixes v, negates every vector perpendicular to v, and satisfies $\rho_{[v]}^2 = I$. It is the unique 3×3 matrix which fixes v and negates every vector perpendicular to v, and its fixed vectors are exactly the vectors of $[v]$ together with 0. Consequently distinct non-null lines have distinct half turns.*

Proof. The first assertions follow from the reflections theorem after multiplication by $-I$, which preserves the dot product, using $\det(-\sigma_v) = (-1)^3 \det \sigma_v = 1$. Since $\mathbb{F}^3 = [v] \oplus v^\perp$, a matrix is determined by its action on v and on $v^\perp$, which gives uniqueness. Writing any vector as $w = \lambda v + w'$ with $w' \in v^\perp$, we have $w\rho_{[v]} = \lambda v - w'$, which equals w precisely when $2w' = 0$, that is $w' = 0$. So the line $[v]$ is recovered from $\rho_{[v]}$. ■

Example 29 *For the same* $v = (2,1,1)$*,*

$$\rho_{[v]} = -\sigma_v = \begin{pmatrix} 3 & 2 & 2 \\ -2 & -2 & -1 \\ -2 & -1 & -2 \end{pmatrix} \in SO(1,2),$$

fixing v *and negating perpendicular vectors:* $v\rho_{[v]} = v$*,* $(1,1,1)\,\rho_{[v]} = (-1,-1,-1)$*. Rows have quadrances* $1,-1,-1$*, trace* -1*, and half spread* $p = (3-(-1))/4 = 1$*.*

Theorem 30 (Two reflections theorem) *Every rotation* $\rho \in SO(1,2,\mathbb{F})$ *is a product* $\rho = \sigma_{v_1}\sigma_{v_2}$ *of two reflections in non-null vectors, and conversely every such product is a rotation. The factorization is not unique.*

The converse is immediate: a product of two elements of $O(1,2)$ each of determinant -1 lies in $SO(1,2)$. The direct statement is proved in Section 7.2, where the factorization is constructed from the anchor of ρ and its freedom is made precise.

Theorem 31 (Axis of a composition) *If* v_1, v_2 *are non-null and independent, then* $\sigma_{v_1}\sigma_{v_2}$ *fixes every vector on the line* $[v_1 \times v_2]$*.*

Proof. $v_1 \times v_2$ is perpendicular to both v_1 and v_2, so it is fixed by both reflections. ■

Example 32 *Let* $v_1 = (1,2,3)$*,* $v_2 = (2,-1,1)$*, with quadrances* $-12, 2$ *and dot product* $\langle v_1, v_2\rangle = 1$*. Then*

$$\sigma_{v_1} = \frac{1}{6}\begin{pmatrix} 7 & 2 & 3 \\ -2 & 2 & -6 \\ -3 & -6 & -3 \end{pmatrix}, \qquad \sigma_{v_2} = \begin{pmatrix} -3 & 2 & -2 \\ -2 & 2 & -1 \\ 2 & -1 & 2 \end{pmatrix},$$

and

$$\sigma_{v_1}\sigma_{v_2} = \frac{1}{6}\begin{pmatrix} -19 & 15 & -10 \\ -10 & 6 & -10 \\ 15 & -15 & 6 \end{pmatrix} \in SO(1,2),$$

with fixed vector $v_1 \times v_2 = (-5,5,-5)$*, i.e. the line* $[(1,-1,1)]$*: indeed* $(1,-1,1)\,\sigma_{v_1}\sigma_{v_2} = (1,-1,1)$*.*

5 The cap of an anchor

Definition 33 *The* ***cap*** *of an anchor* A *is*

$$c(A) \equiv (I+A)(I-A)^{-1}$$

whenever $I - A$ *is invertible.*

This is a form of the classical Cayley transform [3]. The characteristic polynomial $X^3 + Q(v)X$ at $X = \pm 1$ gives

$$\det(I - A_v) = \det(I + A_v) = 1 + Q(v),$$

so the cap of A_v is defined precisely when $Q(v) \neq -1$. Over $\mathbb{Q}$ this excluded set is a non-empty quadric containing every unit spacelike direction. Projectivization in Section 11 places it in its natural setting.

Lemma 34 *If $Q \equiv Q(v) \neq -1$ and $A \equiv A_v$, then*

$$(I - A)^{-1} = I + \frac{A + A^2}{1 + Q}.$$

Proof. Using $A^3 = -QA$ from the characteristic polynomial,

$$(I - A)\left((1 + Q)I + A + A^2\right) = (1 + Q)I + A + A^2 - (1 + Q)A - A^2 - A^3 = (1 + Q)I.$$

■

Theorem 35 (Quadratic reduction) *If $Q \equiv Q(v) \neq -1$ and $A \equiv A_v$, then*

$$c(A) = I + \frac{2}{1 + Q}\left(A + A^2\right).$$

Proof. By the lemma,

$$c(A) = (I + A)\left(I + \frac{A + A^2}{1 + Q}\right) = I + A + \frac{A + 2A^2 + A^3}{1 + Q} = I + A + \frac{(1 - Q)A + 2A^2}{1 + Q} = I + \frac{2(A + A^2)}{1 + Q}.$$

■

In coordinates, with $v = (t, r, s)$, $Q = t^2 - r^2 - s^2$:

$$c(A_v) = \frac{1}{1 + Q}\begin{pmatrix} t^2 + r^2 + s^2 + 1 & 2(s + tr) & 2(ts - r) \\ 2(s - tr) & 1 - t^2 - r^2 + s^2 & 2(t - rs) \\ -2(r + st) & -2(t + rs) & 1 - t^2 + r^2 - s^2 \end{pmatrix}.$$

Theorem 36 (Caps are rotations) *If $Q(v) \neq -1$ and $\rho \equiv c(A_v)$, then:*

1. *$\rho \in SO(1, 2)$;*
2. *$v\rho = v$;*
3. *the characteristic polynomial of ρ is*

$$(X - 1)\left(X^2 - 2\frac{1 - Q}{1 + Q}X + 1\right);$$

4. *-1 is not an eigenvalue of ρ, so no cap is a half turn;*
5. *the cap is injective: $c(A)$ determines A.*

Proof. For 1, the factors $I \pm A$ and their inverses are commuting polynomials in A, and $(I + A)^\dagger = I - A$, so

$$\rho^\dagger \rho = (I - A)(I + A)^{-1}(I + A)(I - A)^{-1} = I,$$

the isometry condition, while $\det \rho = \det(I + A) / \det(I - A) = 1$. For 2, $vA = 0$. For 3, $\operatorname{tr} \rho = 3 + \frac{2}{1+Q}(\operatorname{tr} A + \operatorname{tr} A^2) = 3 - \frac{4Q}{1+Q} = \frac{3-Q}{1+Q}$, using $\operatorname{tr} A^2 = -2Q$. For 4, the quadratic factor at $X = -1$ is $\frac{4}{1+Q} \neq 0$, while a half turn negates the non-zero plane $v^\perp$. For 5, the map $w \mapsto (I + w)(I - w)^{-1}$ is inverted by $z \mapsto (z - I)(z + I)^{-1}$, defined on ρ by 4. ■

Example 37 *Take $v = (0, 1, 1)$, spacelike with $Q = -2$, so $2/(1 + Q) = -2$. Then*

$$A_v = \begin{pmatrix} 0 & 1 & -1 \\ 1 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \qquad A_v^2 = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & -1 & 1 \end{pmatrix},$$

and

$$c(A_v) = I - 2\left(A_v + A_v^2\right) = \begin{pmatrix} -3 & -2 & 2 \\ -2 & -1 & 2 \\ 2 & 2 & -1 \end{pmatrix}.$$

Rows have quadrances $1, -1, -1$, and $(0, 1, 1)\, c(A_v) = (0, 1, 1)$.

Example 38 (A spatial rotation) *$v = (1, 0, 0)$ is timelike with $Q = 1$, and*

$$c(A_v) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}$$

is the quarter turn of the spatial (r, s) plane, fixing the time axis. Purely spatial rotations embed in the relativistic theory with Euclidean arithmetic intact.

Example 39 (A boost) *The unit spacelike vector $v = (0, 0, 1)$ lies on the excluded affine quadric $Q = -1$, so it is not itself an anchor of a cap. The same fixed line may be represented by the anchor vector $\frac{1}{3}v$, with $Q = -1/9$, and*

$$c\left(A_{(0,0,1/3)}\right) = \frac{1}{4}\begin{pmatrix} 5 & 3 & 0 \\ 3 & 5 & 0 \\ 0 & 0 & 4 \end{pmatrix}$$

is the Lorentz boost of velocity $3/5$: it preserves $t^2 - r^2 - s^2$, its rows have quadrances $1, -1, -1$, and its half spread is $p = -1/8$. In the proper orthochronous component, negative half spread is the boost range (Section 12.1).

Example 40 (A parabolic rotation) *The null anchor has a cap, and it is unipotent: if $Q(v) = 0$, then $c(A_v) = I + 2(A_v + A_v^2)$ with characteristic polynomial $(X - 1)^3$. For $v = (1, 1, 0)$,*

$$c(A_v) = \begin{pmatrix} 3 & 2 & -2 \\ -2 & -1 & 2 \\ -2 & -2 & 1 \end{pmatrix}$$

is a rotation fixing v, all eigenvalues 1, satisfying $(\rho - I)^3 = 0$, not the identity. These are the null rotations of relativistic physics; in the Euclidean theory they exist only over finite fields.

Part 4 identifies one limitation of the cap: its image omits every half turn. The excluded quadric $Q = -1$ is a second, on the domain side. Both, and more, are removed in Section 11; until then we develop the affine theory, where the computational content lies.

6 Half spreads

Definition 41 *The **half spread** of a rotation $\rho \in SO(1,2)$ is*

$$p(\rho) \equiv \frac{3 - \operatorname{tr}\rho}{4}.$$

Theorem 42 (Half spread identities) *If $\rho = c(A_v)$ with $Q \equiv Q(v) \neq -1$, then $p \equiv p(\rho)$ satisfies*

$$p = \frac{Q}{1+Q}, \qquad 1-p = \frac{1}{1+Q}, \qquad 1-2p = \frac{1-Q}{1+Q}, \qquad p(1-p) = \frac{Q}{(1+Q)^2},$$

and $p \neq 1$, with $Q = p/(1-p)$. The characteristic polynomial of any rotation ρ is

$$(X-1)\left(X^2 - 2(1-2p(\rho))X + 1\right).$$

If ρ is a half turn, then $p(\rho) = 1$.

Proof. For a cap, the trace formula of Section 5 gives $p = \frac{1}{4}\left(3 - \frac{3-Q}{1+Q}\right) = \frac{Q}{1+Q}$, and the remaining identities follow by field arithmetic; $p = 1$ would force $1 = 0$. By the fixed-direction lemma of Section 4, every rotation has $X - 1$ as a factor of its characteristic polynomial. Since $\det\rho = 1$, the remaining quadratic factor has constant term 1, so it is $X^2 - \alpha X + 1$, and comparing X^2 coefficients gives $\alpha = \operatorname{tr}\rho - 1 = 2 - 4p$. A half turn has trace -1 and hence $p = 1$. ∎

Lemma 43 (Half turn lemma) *A rotation $\rho \in SO(1,2,\mathbb{F})$ has $p(\rho) = 1$ precisely when it is a half turn. In that case $\rho = \rho_{[f]}$, where $[f]$ is the fixed line of ρ, and f is non-null.*

Proof. A half turn has $p = 1$ by the half spread identities, so suppose $p(\rho) = 1$. Then the characteristic polynomial of ρ is $(X-1)(X+1)^2$.

Step 1: the fixed line. The root 1 is simple, since $(X+1)^2$ takes the value $4 \neq 0$ at $X = 1$. So the eigenvalue 1 has a one dimensional eigenspace $[f]$, and $f \neq 0$ is fixed by ρ.

Step 2: an invariant complement. Let $W \equiv \left\{w : w(\rho+I)^2 = 0\right\}$. Since ρ commutes with $(\rho+I)^2$, W is invariant under ρ. The polynomial identity

$$(X+1)^2 - (X-1)(X+3) = 4$$

gives, for every vector w,

$$w = \tfrac{1}{4}w(\rho+I)^2 - \tfrac{1}{4}w(\rho-I)(\rho+3I).$$

By the Cayley–Hamilton theorem $(\rho-I)(\rho+I)^2 = 0$, so the first term is fixed by ρ and the second lies in W. If w is both fixed and in W, both terms vanish and $w = 0$. Hence $\mathbb{F}^3 = [f] \oplus W$, and W has dimension two.

Step 3: $W = f^{\perp}$, and f is non-null. Let $w \in W$ and put $g \equiv w(\rho+I)$, so that $g(\rho+I) = 0$, that is $g\rho = -g$, and $w\rho = -w + g$. Since ρ preserves the dot product and fixes f,

$$\langle f, g\rangle = \langle f\rho, g\rho\rangle = -\langle f, g\rangle,$$

so $\langle f, g\rangle = 0$; and then

$$\langle f, w\rangle = \langle f\rho, w\rho\rangle = \langle f, -w + g\rangle = -\langle f, w\rangle,$$

so $\langle f, w\rangle = 0$. Thus $W \subseteq f^{\perp}$. Since the form is non-degenerate and $f \neq 0$, $f^{\perp}$ has dimension two, so $W = f^{\perp}$. As $f \notin W$ by Step 2, $f \notin f^{\perp}$, that is $Q(f) \neq 0$.

Step 4: ρ acts on W as $-I$. Write B for the restriction of the dot product to W. It is non-degenerate: a vector of W perpendicular to all of W is also perpendicular to f, hence to all of $\mathbb{F}^3$, and so is 0. Choose a non-null $e_1 \in W$ (one exists, for if $B(x, x) = 0$ for all $x \in W$ then $B = 0$ by polarization), and let e_2 span the line of W perpendicular to e_1; e_2 is non-null by non-degeneracy. Put $a \equiv B(e_1, e_1)$ and $b \equiv B(e_2, e_2)$, both non-zero.

Let σ be the restriction of $-\rho$ to W. In the basis f, e_1, e_2 the matrix of ρ is block diagonal, so $1 = \det\rho = \det(-\sigma) = \det\sigma$, and σ has characteristic polynomial $(X-1)^2$. By the Cayley–Hamilton theorem $\sigma^2 - 2\sigma + I = 0$, so $N \equiv \sigma - I$ satisfies $N^2 = 0$, and $\sigma^{-1} = 2I - \sigma$. Since σ preserves B,

$$B(x\sigma, y) = B\left(x, y\sigma^{-1}\right) = B(x, y(2I - \sigma)),$$

and subtracting $B(x, y)$ from both sides gives

$$B(xN, y) = -B(x, yN) \qquad \text{for all } x, y \in W.$$

Write $e_iN = n_{i1}e_1 + n_{i2}e_2$. Taking $x = e_i$, $y = e_j$ gives $n_{ij}B(e_j, e_j) = -n_{ji}B(e_i, e_i)$. For $i = j$ this gives $n_{11} = n_{22} = 0$, and for $i = 1, j = 2$ it gives $n_{21} = -\frac{b}{a}n_{12}$. Hence

$$N = \begin{pmatrix} 0 & n_{12} \\ -\frac{b}{a}n_{12} & 0 \end{pmatrix}, \qquad N^2 = -\frac{b}{a}n_{12}^2 I,$$

and $N^2 = 0$ forces $n_{12} = 0$, so $N = 0$. Thus $\sigma = I$, and ρ negates every vector of $W = f^{\perp}$.

Conclusion. ρ fixes the non-null vector f and negates every vector perpendicular to f, so $\rho = \rho_{[f]}$ by the uniqueness in the half turns theorem of Section 4. ■

Over $\mathbb{Q}$ the form $t^2 - r^2 - s^2$ represents every rational number, so every rational $p \neq 1$ occurs as the half spread of a cap, while $p = 1$ occurs for half turns. The causal meaning of the different ranges is given in Section 12.1.

Example 44 *For $\rho = c(A_v)$ with $v = (0, 1, 1)$ and $Q = -2$, the trace is -5, so*

$$p = \frac{3 - (-5)}{4} = 2 = \frac{Q}{1+Q}, \quad 1 - p = -1 = \frac{1}{1+Q}, \quad p(1-p) = -2 = \frac{Q}{(1+Q)^2},$$

and $Q = p/(1-p) = -2$. The characteristic polynomial is $(X-1)(X^2 + 6X + 1)$, since $1 - 2p = -3$.

Definition 45 *For a cap $\rho = c(A_v)$ with v non-null, or a half turn $\rho = \rho_{[v]}$, the **spread** is*

$$s(\rho) \equiv s(w, w\rho)$$

for any non-null $w \perp v$.

Theorem 46 (Spread of a rotation) *The spread is well defined and independent of w, and*

$$s(\rho) = 4p(1-p) \qquad \text{where} \qquad p = p(\rho).$$

Proof. Let $\rho = c(A_v)$, $Q \equiv Q(v)$, and let w be non-null with $\langle w, v\rangle = 0$. Then $wA_v = v \times w$ and $wA_v^2 = w\left(v^\dagger v\right) - Qw = -Qw$, so

$$w\rho = \frac{(1-Q)\,w + 2\,(v \times w)}{1+Q}.$$

Since $\langle v \times w, w\rangle = 0$ and $Q(w\rho) = Q(w)$,

$$s(w, w\rho) = 1 - \frac{(1-Q)^2}{(1+Q)^2} = \frac{4Q}{(1+Q)^2} = 4p(1-p).$$

For a half turn, $w\rho = -w$ and the spread is $0 = 4 \cdot 1 \cdot 0$. Such a w exists because the restriction of the dot product to $v^\perp$ is non-degenerate for non-null v. ∎

The polynomial $S_2(p) = 4p(1-p)$ is the second spread polynomial of rational trigonometry [10]. The parabolic rotations have $p = 0$ without being the identity: over $\mathbb{Q}$, the half spread does not separate the identity from all its neighbours; the full separation is achieved by the anchor.

Example 47 *Continuing with $\rho = c(A_v)$, $v = (0,1,1)$, take $w = (1,1,-1)$, with $\langle w, v\rangle = 0$ and $Q(w) = -1$. Then $w\rho = (-7,-5,5)$, of quadrance -1, and $\langle w, w\rho\rangle = 3$, so*

$$s(\rho) = s(w, w\rho) = 1 - \frac{9}{(-1)(-1)} = -8 = 4p(1-p).$$

Theorem 48 (Spread of normals) *For non-null v_1, v_2,*

$$p(\sigma_{v_1}\sigma_{v_2}) = s(v_1, v_2).$$

Proof. Since $\operatorname{tr}\left(v^\dagger v\right) = Q(v)$ and $\left(v_1^\dagger v_1\right)\left(v_2^\dagger v_2\right) = v_1^\dagger \langle v_1, v_2\rangle v_2$ has trace $\langle v_1, v_2\rangle^2$,

$$\operatorname{tr}(\sigma_{v_1}\sigma_{v_2}) = 3 - 2 - 2 + \frac{4\langle v_1, v_2\rangle^2}{Q(v_1)\,Q(v_2)} = -1 + 4(1 - s(v_1, v_2)),$$

so $p = (3 - \operatorname{tr})/4 = s(v_1, v_2)$. ∎

Example 49 *For $v_1 = (1,2,3)$ and $v_2 = (2,-1,1)$, $s(v_1, v_2) = 1 - 1/((-12)\cdot 2) = 25/24$, and the product computed earlier has trace $(-19+6+6)/6 = -7/6$, giving $p = \frac{1}{4}(3 + 7/6) = 25/24$.*

7 The anchor of a rotation

Definition 50 *The **anchor** of a rotation ρ with $p(\rho) \neq 1$ is*

$$A(\rho) \equiv (\rho - I)(\rho + I)^{-1}.$$

By the characteristic polynomial of Section 6, -1 is an eigenvalue of ρ precisely when the quadratic factor vanishes at $X = -1$, that is when $4(1-p) = 0$. So $\det(I+\rho) \neq 0$ precisely when $p(\rho) \neq 1$, and the anchor is defined exactly there.

Theorem 51 (Anchor of a rotation) *For a rotation ρ with $p \equiv p(\rho) \neq 1$:*

1. $A(\rho)$ lies in $so(1,2)$, so $A(\rho) = A_w$ for a unique vector w;

2. $c(A(\rho)) = \rho$ and $A(c(A)) = A$: the cap and the anchor are mutually inverse bijections between $\{A_w : Q(w) \neq -1\}$ and $\{\rho : p(\rho) \neq 1\}$;

3. $w\rho = w$;

*4. $Q(w) = p/(1-p)$, the **half twist** of ρ;*

5. writing $\rho^{as} \equiv \frac{1}{2}\left(\rho - \rho^{\dagger}\right)$ for the adjoint-antisymmetric part,

$$A(\rho) = \frac{2}{1+\operatorname{tr}\rho}\rho^{as}.$$

Proof. For 1, since $\rho^{\dagger} = \rho^{-1}$,

$$A(\rho)^{\dagger} = \left(\rho^{-1} - I\right)\left(\rho^{-1} + I\right)^{-1} = (I-\rho)\rho^{-1} \cdot \rho(I+\rho)^{-1} = -A(\rho).$$

For 2, the two Cayley-type maps are mutually inverse wherever defined, and part 4 of the theorem that caps are rotations shows the domains match: $Q(w) = p/(1-p) = -1$ would force $0 = -1$, so anchors of rotations avoid the excluded quadric. For 5, writing $\rho = I + \frac{2}{1+Q}(A_w + A_w^2)$, the term $A_w^2 = w^{\dagger}w - QI$ is self-adjoint, so

$$\rho^{as} = \frac{2}{1+Q}A_w, \qquad 1 + \operatorname{tr}\rho = \frac{4}{1+Q},$$

and dividing gives the formula; 3 and 4 follow from Section 5 and the half spread identities. ∎

Part 5 computes the anchor by adjoint-antisymmetrization and one division, with no matrix inversion. Thus the anchor is recovered directly from the rotation matrix.

Example 52 *We recover the axis of $\rho = c(A_v)$, $v = (0,1,1)$, from ρ alone. With $\operatorname{tr}\rho = -5$ the adjoint-antisymmetric part is*

$$\rho^{as} = \frac{1}{2}\left(\rho - \rho^{\dagger}\right) = \begin{pmatrix} 0 & -2 & 2 \\ -2 & 0 & 0 \\ 2 & 0 & 0 \end{pmatrix},$$

and

$$A(\rho) = \frac{2}{1+\operatorname{tr}\rho}\rho^{as} = -\frac{1}{2}\rho^{as} = \begin{pmatrix} 0 & 1 & -1 \\ 1 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix} = A_{(0,1,1)},$$

so the anchor vector is $w = (0,1,1)$, with half twist $Q(w) = -2 = p/(1-p)$.

7.1 The anchor of a product of two reflections

Theorem 53 (Anchor of two reflections) *If v_1, v_2 are non-null with $d \equiv \langle v_1, v_2 \rangle \neq 0$, then*

$$A\left(\sigma_{v_1}\sigma_{v_2}\right) = \frac{v_1^\dagger v_2 - v_2^\dagger v_1}{\langle v_1, v_2 \rangle} = \frac{1}{d} A_{v_1 \times v_2},$$

depending only on $[v_1]$ and $[v_2]$, with quadrance $s/(1-s)$ where $s = s(v_1, v_2)$.

Proof. Write $Q_1 \equiv Q(v_1)$ and $Q_2 \equiv Q(v_2)$. Expanding $\rho \equiv \sigma_{v_1}\sigma_{v_2}$, the terms $v_i^\dagger v_i$ are self-adjoint, and the remaining term is $\frac{4d}{Q_1 Q_2} v_1^\dagger v_2$, so by part 2 of the corollary of Section 3,

$$\rho^{as} = \frac{2d}{Q_1 Q_2}\left(v_1^\dagger v_2 - v_2^\dagger v_1\right) = \frac{2d}{Q_1 Q_2} A_{v_1 \times v_2},$$

while by the spread of normals theorem $1 + \operatorname{tr}\rho = 4(1-s) = \frac{4d^2}{Q_1 Q_2}$. The formula follows from part 5 above, and the quadrance is $Q(v_1 \times v_2)/d^2 = (Q_1 Q_2 - d^2)/d^2 = s/(1-s)$ by the Lagrange identity. ■

Example 54 *For $v_1 = (1,2,3)$ and $v_2 = (2,-1,1)$, we have $d = 1$ and $v_1 \times v_2 = (-5,5,-5)$. Hence*

$$A\left(\sigma_{v_1}\sigma_{v_2}\right) = A_{(-5,5,-5)}.$$

The anchor line is $[(1,-1,1)]$, and

$$\frac{Q(-5,5,-5)}{d^2} = -25 = \frac{s}{1-s}, \qquad s = \frac{25}{24}.$$

7.2 Proof of the two reflections theorem

We now prove the two reflections theorem of Section 4, in the following sharper form: if $\rho \neq I$, the first mirror line may be chosen freely among the non-null lines perpendicular to the anchor vector of ρ, or to the axis of ρ when ρ is a half turn, and the second mirror line is then determined.

Proof. The identity is $\sigma_v \sigma_v$ for any non-null v. In every case the second mirror is determined by the first, since $\sigma_{v_1}\sigma_{v_2} = \rho$ forces $\sigma_{v_2} = \sigma_{v_1}\rho$, and a reflection determines its line by part 2 of the reflections theorem.

If $p(\rho) = 1$, then ρ is a half turn $\rho_{[v]}$ by the half turn lemma of Section 6; choose any non-null v_1 in the plane $v^\perp$, and let v_2 span the line of $v^\perp$ perpendicular to v_1. The restriction of the dot product to $v^\perp$ is non-degenerate, so v_2 is non-null: otherwise it would be perpendicular to all of $v^\perp$. Now $\sigma_{v_1}\sigma_{v_2}$ negates v_1 and v_2, hence every vector of $v^\perp$, and fixes v, which is perpendicular to both mirrors; by the uniqueness in the half turns theorem it equals $\rho_{[v]}$.

If $p(\rho) \neq 1$, then $\rho = c(A_w)$ with $Q \equiv Q(w) \neq -1$ and $w \neq 0$. Choose any non-null v_1 with $\langle v_1, w \rangle = 0$, and set

$$v_2 \equiv v_1(I + A_w) = v_1 + w \times v_1.$$

Then $\langle v_1, v_2 \rangle = Q(v_1) \neq 0$ since $w \times v_1 \perp v_1$; and $Q(v_2) = Q(v_1) + Q(w \times v_1) = Q(v_1)(1+Q) \neq 0$ by the Lagrange identity, so v_2 is non-null; and

$$v_1 \times v_2 = v_1 \times (w \times v_1) = Q(v_1)\, w - \langle v_1, w \rangle v_1 = Q(v_1)\, w$$

by the vector triple product identity. Hence by the anchor of two reflections theorem $A(\sigma_{v_1}\sigma_{v_2}) = \frac{1}{Q(v_1)}A_{Q(v_1)w} = A_w$, and since the anchor is a complete invariant, $\sigma_{v_1}\sigma_{v_2} = \rho$. ■

When w is null, the plane $w^\perp$ is degenerate and contains w. The factorization still exists, but its perpendicularity geometry is degenerate.

Example 55 *Factor $\rho = c(A_w)$ with $w = (0,1,1)$ into two reflections. Choose $v_1 = (1,1,-1)$ perpendicular to w, and set*

$$v_2 \equiv v_1 + w \times v_1 = (1,1,-1) + (2,1,-1) = (3,2,-2),$$

non-null with $Q(v_2) = 1 = Q(v_1)(1 + Q(w))$. Then $\langle v_1, v_2\rangle = -1$ and $v_1 \times v_2 = (0,-1,-1) = Q(v_1)\, w$, so $\sigma_{v_1}\sigma_{v_2}$ has anchor vector w and equals ρ.

8 Algebraic orientation

A rotation and its inverse have the same axis and half spread. Their anchors have opposite signs:

$$A\left(\rho^{-1}\right) = -A(\rho).$$

For timelike axes this sign has the usual orientation interpretation. For spacelike and null axes the Lorentz metric changes the picture.

Theorem 56 (Orientation determinant) *Let $\rho = c(A_v)$ with $Q = Q(v) \neq -1$, and let $w \perp v$. Then*

$$\det\begin{pmatrix} v \\ w \\ w\rho \end{pmatrix} = -\frac{2Q}{1+Q}Q(w).$$

Proof. For $w \perp v$,

$$w\rho = \frac{(1-Q)\, w + 2(v \times w)}{1+Q}.$$

The term proportional to w repeats the second row and contributes zero to the determinant. Hence, cycling the rows and using the scalar triple product identity,

$$\det\begin{pmatrix} v \\ w \\ w\rho \end{pmatrix} = \frac{2}{1+Q}\det\begin{pmatrix} v \\ w \\ v \times w \end{pmatrix} = -\frac{2}{1+Q}Q(v \times w).$$

Since $\langle v, w\rangle = 0$, Lagrange's identity gives $Q(v \times w) = Q(v)\, Q(w)$. ■

Corollary 57 (Timelike orientation) *Over an ordered field, let v be timelike and $w \neq 0$ with $w \perp v$. Then*

$$\det\begin{pmatrix} v \\ w \\ w\rho \end{pmatrix} > 0$$

for $\rho = c(A_v)$.

Proof. Here $Q(v) > 0$, so $1+Q(v) > 0$, while the restriction of the form to $v^{\perp}$ is negative definite, so $Q(w) < 0$. Hence $-\frac{2Q(v)}{1+Q(v)}Q(w) > 0$. ■

Thus, once the cross-product convention and ambient orientation have been fixed, the sense of an ordinary spatial rotation is detected by the sign of one determinant.

For a spacelike axis, $v^{\perp}$ has signature $(1,1)$. The same determinant can therefore have either sign, according to the causal type of w. There is no single Euclidean right-hand test for a boost. For a null axis, $Q(v) = 0$ and the determinant vanishes identically. The two inverse parabolic transformations are still distinguished by the full anchors A_v and A_{-v}, but not by this metrical determinant.

Example 58 *For the boost with anchor $v = (0, 0, 1/3)$, take first $w = (1, 0, 0)$. The determinant is $1/4$. For $w = (0, 1, 0)$ it is $-1/4$. The sign change reflects the indefinite metric on the plane perpendicular to the fixed spacelike axis.*

The invariant sign choice is therefore best attached to the anchor ray rather than described as a universal right-hand rule; we return to this in Section 12.3.

9 The anchor product

Rotations compose; we compute the composition on anchors. Rotations act on row vectors, so $\rho_1\rho_2$ denotes the matrix product, performing ρ_1 first.

Definition 59 *For vectors v, u with $\langle v, u\rangle \neq 1$, the **anchor product** is*

$$v \otimes u \equiv \frac{v + u - v \times u}{1 - \langle v, u\rangle},$$

and correspondingly for anchors, using the anchor form of Section 3,

$$A \otimes B \equiv \frac{A + B + (AB - BA)}{1 - A \cdot B},$$

so that $A_v \otimes A_u = A_{v\otimes u}$, by part 2 of the corollary of Section 3.

Lemma 60 (Quadrance of an anchor product) *For any vectors v and u,*

$$(1 - \langle v, u\rangle)^2 + Q(v + u - v \times u) = (1 + Q(v))(1 + Q(u)).$$

Consequently if $\langle v, u\rangle \neq 1$ then $1 + Q(v \otimes u) = \frac{(1+Q(v))(1+Q(u))}{(1-\langle v,u\rangle)^2}$.

Proof. Write $d \equiv \langle v, u\rangle$. Since $v \times u$ is perpendicular to v and u,

$$Q(v + u - v \times u) = Q(v) + Q(u) + 2d + Q(v \times u) = Q(v) + Q(u) + 2d + Q(v)\,Q(u) - d^2$$

by the Lagrange identity, and adding $(1 - d)^2 = 1 - 2d + d^2$ gives $1 + Q(v) + Q(u) + Q(v)\,Q(u)$. ■

So if the caps of A_v and A_u are defined and $\langle v, u\rangle \neq 1$, the cap of $A_{v\otimes u}$ is also defined.

Theorem 61 (Composition) *If $Q(v) \neq -1$, $Q(u) \neq -1$ and $\langle v, u\rangle \neq 1$, then*

$$c(A_v)\, c(A_u) = c(A_{v\otimes u}).$$

Proof. Write $\rho_1 \equiv c(A_v)$, $\rho_2 \equiv c(A_u)$, $d \equiv \langle v, u\rangle$ and $m \equiv v \times u$. We first treat the case where m is non-null, using reflections, and then use the vanishing lemma for the general case.

Suppose $Q(m) \neq 0$, and set

$$a \equiv m(I - A_v) = m - v \times m, \qquad b \equiv m(I + A_u) = m + u \times m.$$

Since $m \perp v$ and $m \perp u$, the construction of Section 7.2, applied to $\rho_1^{-1} = c(A_{-v})$ with first mirror m, gives $\sigma_m\sigma_a = \rho_1^{-1}$, that is $\rho_1 = \sigma_a\sigma_m$; applied to ρ_2 with first mirror m it gives $\rho_2 = \sigma_m\sigma_b$. (The mirrors are non-null: by the Lagrange identity $Q(a) = Q(m)(1 + Q(v))$ and $Q(b) = Q(m)(1 + Q(u))$, all factors non-zero.) The shared middle mirror cancels:

$$\rho_1\rho_2 = \sigma_a\sigma_m\sigma_m\sigma_b = \sigma_a\sigma_b.$$

By Binet–Cauchy, using $\langle v, m\rangle = \langle u, m\rangle = 0$,

$$\langle a, b\rangle = Q(m) - \langle v \times m, u \times m\rangle = Q(m) - dQ(m) = (1 - d)\, Q(m) \neq 0,$$

so the anchor of two reflections theorem applies. By the vector triple product identity, $m \times (u \times m) = Q(m)\, u$ and $m \times (v \times m) = Q(m)\, v$, while

$$(v \times m) \times (u \times m) = \langle v \times m, m\rangle\, u - \langle v \times m, u\rangle\, m = -\langle v \times m, u\rangle\, m = Q(m)\, m,$$

since by the scalar triple product identity $\langle v \times m, u\rangle = -\det(v, m, u) = \det(v, u, m) = -\langle v \times u, m\rangle = -Q(m)$. Hence

$$a \times b = Q(m)\, u + Q(m)\, v - Q(m)\, m = Q(m)(v + u - v \times u),$$

and therefore

$$A(\rho_1\rho_2) = \frac{1}{\langle a, b\rangle} A_{a\times b} = A_{\frac{v+u-v\times u}{1-d}} = A_{v\otimes u}.$$

Since the anchor is a complete invariant, $\rho_1\rho_2 = c(A_{v\otimes u})$.

For the general case write $n \equiv v + u - v \times u$, so that $v \otimes u = \frac{1}{1-d} n$. By quadratic reduction and the lemma,

$$(1 + Q(v))(1 + Q(u))\, c(A_{v\otimes u}) = (1 + Q(v))(1 + Q(u))\, I + 2\left((1 - d)\, A_n + A_n^2\right),$$

while

$$(1 + Q(v))(1 + Q(u))\, c(A_v)\, c(A_u) = \left((1 + Q(v))\, I + 2\left(A_v + A_v^2\right)\right)\left((1 + Q(u))\, I + 2\left(A_u + A_u^2\right)\right).$$

The difference P of the two right hand sides is a matrix of polynomials with integer coefficients in the six coordinates of v and u, and the theorem asserts precisely that $P = 0$. By the first part of the proof, P vanishes at every rational point where

$$h \equiv Q(v \times u)(1 - d)(1 + Q(v))(1 + Q(u))$$

does not vanish, and h is not the zero polynomial. By the vanishing lemma of Section 2, P is the zero polynomial, so $P = 0$ over every field. ■

The anchor product carries $-v \times u$, as in the Euclidean theory, because both theories compose matrices acting on row vectors, first factor first. The same formula with $+v \times u$ is the **opposite product** $v \otimes^{op} u = u \otimes v$, composing in reverse order.

Example 62 *Let* $v = (2,1,0)$, $u = (1,0,2)$, *so* $Q(v) = 3$, $Q(u) = -3$, $d = 2$, $v \times u = (-2,-4,-1)$. *Then*

$$v \otimes u = \frac{(3,1,2) - (-2,-4,-1)}{1-2} = (-5,-5,-3),$$

so the product of the rotations with anchor vectors $(2,1,0)$, $(1,0,2)$, *of half spreads* $3/4$ *and* $3/2$, *is the rotation with anchor vector* $(-5,-5,-3)$, *of half spread* $9/8$. *As the lemma predicts,* $(1-d)^2 (1 + Q(-5,-5,-3)) = 1 \cdot (-8) = (1+3)(1-3)$.

Corollary 63 (Half turn criterion) *For* $Q(v), Q(u) \neq -1$, *the product* $c(A_v)\, c(A_u)$ *is a half turn precisely when* $\langle v, u \rangle = 1$, *and in that case it is the half turn about* $[v + u - v \times u]$.

Proof. If $\langle v, u \rangle \neq 1$ the product is a cap by the composition theorem, so -1 is not an eigenvalue, while every half turn negates a non-zero plane; so the product is not a half turn. If $d \equiv \langle v, u \rangle = 1$, write $n \equiv v + u - v \times u$. The lemma gives $Q(n) = (1 + Q(v))(1 + Q(u)) \neq 0$, and the polynomial identity $P = 0$ from the proof of the composition theorem, divided by this quantity, reads

$$c(A_v)\, c(A_u) = I + \frac{2A_n^2}{Q(n)} = -I + \frac{2}{Q(n)} n^\dagger n = \rho_{[n]},$$

using $A_n^2 = n^\dagger n - Q(n)\, I$. ■

Example 64 *The spatial quarter turn* $c\left(A_{(1,0,0)}\right)$ *has* $\langle (1,0,0), (1,0,0) \rangle = 1$, *so its square is a half turn:*

$$c\left(A_{(1,0,0)}\right)^2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}^2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix} = \rho_{[(1,0,0)]},$$

the half turn about the time axis, in agreement with $n = (2,0,0)$.

Corollary 65 (Tangent addition) *For a fixed non-null vector* a *with* $Q \equiv Q(a)$, *the caps* $c(xA_a)$ *compose by*

$$c(xA_a)\, c(yA_a) = c(zA_a), \qquad z = \frac{x+y}{1-Qxy},$$

whenever the caps on the left are defined and $Qxy \neq 1$: *the one-parameter family through a given rotation composes by the rational addition law with modulus* Q.

Proof. For $v = xa$ and $u = ya$ we have $v \times u = 0$ and $\langle v, u \rangle = Qxy$, so $v \otimes u = \frac{x+y}{1-Qxy} a$. ■

Example 66 *Take $a = (1,0,0)$, $Q(a) = 1$. The cap*

$$c\left(\tfrac{1}{3}A_a\right) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 4/5 & 3/5 \\ 0 & -3/5 & 4/5 \end{pmatrix}$$

is the $3,4,5$ spatial rotation. Composed with itself, $z = \frac{1/3+1/3}{1-(1/3)(1/3)} = 3/4$, so

$$c\left(\tfrac{1}{3}A_a\right) c\left(\tfrac{1}{3}A_a\right) = c\left(\tfrac{3}{4}A_a\right) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 7/25 & 24/25 \\ 0 & -24/25 & 7/25 \end{pmatrix},$$

the $7,24,25$ rotation. For a timelike axis the tangent addition law is the ordinary circular one.

Corollary 67 (Collinear boost anchors) *Let e be spacelike with $Q(e) = -1$. For anchor vectors ae and be,*

$$(ae) \otimes (be) = \frac{a+b}{1+ab}\, e$$

whenever $ab \neq -1$.

Proof. The cross product vanishes for parallel vectors and $\langle ae, be\rangle = -ab$. ∎

The parameters a and b are half-rapidity parameters, not physical velocities. Over $\mathbb{R}$ put

$$\beta(a) \equiv \frac{2a}{1+a^2}.$$

Then direct substitution gives

$$\beta\left(\frac{a+b}{1+ab}\right) = \frac{\beta(a)+\beta(b)}{1+\beta(a)\beta(b)}.$$

Thus the Einstein velocity addition law [5] is a consequence of the anchor composition law.

Example 68 *The boost of Section 5 has anchor parameter $a = 1/3$, so its physical velocity is*

$$\beta = \frac{2/3}{1+1/9} = \frac{3}{5}.$$

Composing this boost with itself gives anchor parameter

$$a_{12} = \frac{1/3+1/3}{1+1/9} = \frac{3}{5},$$

and hence physical velocity

$$\beta_{12} = \frac{2\,(3/5)}{1+9/25} = \frac{15}{17}.$$

Equivalently,

$$\frac{3/5+3/5}{1+9/25} = \frac{15}{17}.$$

Example 69 (Two non-collinear boosts) *Let*

$$v = \left(0, 0, \frac{1}{3}\right), \qquad u = \left(0, \frac{1}{3}, 0\right).$$

These are anchors of boosts in two perpendicular spatial directions; anchors of pure boosts relative to the time axis $(1, 0, 0)$ *have first coordinate* 0*. Since* $\langle v, u\rangle = 0$ *and*

$$v \times u = \left(\frac{1}{9}, 0, 0\right),$$

their product has anchor

$$v \otimes u = \left(-\frac{1}{9}, \frac{1}{3}, \frac{1}{3}\right).$$

Reversing the order gives

$$u \otimes v = \left(\frac{1}{9}, \frac{1}{3}, \frac{1}{3}\right).$$

The time component, produced entirely by the cross product, is absent from both factors and changes sign when the boosts are reversed. This is the algebraic source of the Thomas–Wigner rotation for non-collinear boosts.

Theorem 70 (Conditional associativity) *The anchor product is associative wherever all products occurring are defined: if* $v \otimes u$*,* $(v \otimes u) \otimes w$*,* $u \otimes w$ *and* $v \otimes (u \otimes w)$ *all exist, and the caps of* A_v*,* A_u *and* A_w *are defined, then*

$$(v \otimes u) \otimes w = v \otimes (u \otimes w).$$

It is genuinely partial: one side may exist while the other does not.

Proof. When all four products exist, both sides are anchor vectors of the single rotation $c(A_v)\, c(A_u)\, c(A_w)$ by the composition theorem, and anchors are unique. Partiality is shown by the following example. ∎

Example 71 *Let* $v = (2, 0, 0)$*,* $u = (1, 2, 3)$*,* $u' = (2, -1, 1)$*. The right grouping cannot form:* $\langle u, u'\rangle = 1$*, so* $c(A_u)\, c(A_{u'})$ *is a half turn and* $u \otimes u'$ *does not exist. The left grouping forms:* $\langle v, u\rangle = 2$ *gives* $v \otimes u = (-3, -8, 1)$*, and* $\langle v \otimes u, u'\rangle = -15 \neq 1$*, so* $(v \otimes u) \otimes u'$ *is defined. One association exists while the other does not.*

10 Four breakdowns of the affine theory

The dictionary between rotations and anchors is complete except at four points.

First breakdown: half turns have no anchors. The cap misses the rotations with $p = 1$, a family parametrized by the non-null lines of $\mathbb{F}^3$—a projective plane with its null conic removed, and over $\mathbb{Q}$ that conic is present.

Second breakdown: the anchor product has a forbidden locus. The product $v \otimes u$ fails precisely when $\langle v, u\rangle = 1$, and by the half turn criterion the composed rotation is a half turn, so there is no anchor for the product to be.

Example 72 *Return to* $v_1 = (1,2,3)$, $v_2 = (2,-1,1)$, *whose dot product is* 1. *The rotations* $c(A_{v_1})$, $c(A_{v_2})$ *have half spreads* $12/11$ *and* $2/3$, *yet the numerator of the anchor product survives the death of its denominator:*

$$v_1 + v_2 - v_1 \times v_2 = (3,1,4) - (-5,5,-5) = (8,-4,9),$$

and indeed

$$c(A_{v_1})\, c(A_{v_2}) = \frac{1}{33} \begin{pmatrix} -161 & 64 & -144 \\ -64 & -1 & -72 \\ 144 & -72 & 129 \end{pmatrix} = \rho_{[(8,-4,9)]},$$

the half turn about $(8,-4,9)$, *of trace* -1. *The quadrance of an anchor product lemma confirms the product:*

$$(1-12)(1+2) = -33 = 0^2 + Q(8,-4,9).$$

The repair: introduce x and consider $c\left(\frac{1}{x}A\right)$ as a matrix of rational functions of x, defined at $x = 0$ where it equals the half turn. The pair $(x : v)$ parametrizes uniformly what the anchor vector $\frac{1}{x}v$ parametrizes only for $x \neq 0$, and the half turn is an **anchor at infinity**.

Third breakdown: associativity is conditional. The group $SO(1,2)$ is associative; the defect belongs to the parametrization.

Fourth breakdown: an excluded quadric of anchors. The anchors with $Q(v) = -1$ have no cap and correspond to no rotation; over $\mathbb{Q}$ they form a non-empty quadric containing every unit spacelike direction. In the Euclidean theory this breakdown is invisible over $\mathbb{Q}$ and appears only over finite fields; here it is rational, and it is why the boost of Section 5 is anchored at $\frac{1}{3}(0,0,1)$ rather than $(0,0,1)$.

The four breakdowns share a single cause and a single cure. The anchor vector v is the affine chart $x = 1$ of a projective object, the proportion $(x : v)$; the half turns are its points at infinity; the forbidden locus is where a product lands at infinity; conditional associativity is what associativity looks like through a chart that cannot see infinity; and the excluded quadric is the affine trace of a projective quadric of proportions corresponding to no rotation. In the next section we abandon the chart.

11 Dihedrons, matrices and the projective rotation group

The affine anchor is one chart of a projective object. In the Minkowski case the projective algebra is especially simple: it is the full algebra of 2×2 matrices over the ground field.

Definition 73 *A* ***dihedron*** *over* $\mathbb{F}$ *is a vector*

$$\alpha = (x, v) = (x, t, r, s) \in \mathbb{F}^4,$$

with ***scalar part*** x *and* ***vector part*** v; *we identify a vector* $w \in \mathbb{F}^3$ *with the pure dihedron* $(0, w)$ *and a scalar* x *with* $(x, 0)$. *Its* ***matrix*** *is*

$$\mu(\alpha) \equiv \begin{pmatrix} x+s & r-t \\ r+t & x-s \end{pmatrix}.$$

*Its **conjugate** is*

$$\overline{\alpha} \equiv (x, -t, -r, -s)\,,$$

*and its **quadrance** is*

$$N(\alpha) \equiv x^2 + t^2 - r^2 - s^2 = x^2 + Q(v)\,.$$

Thus

$$\operatorname{tr}\mu(\alpha) = 2x, \qquad \det\mu(\alpha) = N(\alpha)\,.$$

A dihedron is **null** precisely when its matrix is singular.

Write $\mathbf{i} \equiv (0,1,0,0)$, $\mathbf{j} \equiv (0,0,1,0)$, $\mathbf{k} \equiv (0,0,0,1)$, so that $(x,t,r,s) = x + t\mathbf{i} + r\mathbf{j} + s\mathbf{k}$, and

$$J \equiv \mu(\mathbf{i}) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad K \equiv \mu(\mathbf{j}) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad D \equiv \mu(\mathbf{k}) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Theorem 74 (Dihedron matrix algebra) *The map*

$$\mu : \mathbb{F}^4 \longrightarrow M_2(\mathbb{F})$$

is a linear bijection. Pulling matrix multiplication back through μ gives the product

$$(x, v) \otimes (y, u) = (xy - \langle v, u\rangle\,, xu + yv - v \times u)\,,$$

determined by bilinearity and the table

$$\mathbf{i}\otimes\mathbf{i} = -1, \qquad \mathbf{j}\otimes\mathbf{j} = \mathbf{k}\otimes\mathbf{k} = 1, \qquad \mathbf{j}\otimes\mathbf{i} = \mathbf{k}, \quad \mathbf{j}\otimes\mathbf{k} = \mathbf{i}, \quad \mathbf{i}\otimes\mathbf{k} = \mathbf{j},$$

with $\mathbf{i}\otimes\mathbf{j} = -\mathbf{k}$, $\mathbf{k}\otimes\mathbf{j} = -\mathbf{i}$, $\mathbf{k}\otimes\mathbf{i} = -\mathbf{j}$. With this product:

$$\mu(\alpha\otimes\beta) = \mu(\alpha)\,\mu(\beta)\,, \qquad \mu(\overline{\alpha}) = \operatorname{adj}\mu(\alpha)\,, \qquad \alpha\otimes\overline{\alpha} = \overline{\alpha}\otimes\alpha = N(\alpha)\,,$$

and

$$N(\alpha\otimes\beta) = N(\alpha)\,N(\beta)\,.$$

Consequently the product is associative, and a dihedron is invertible precisely when it is non-null, with

$$\alpha^{-1} = \frac{\overline{\alpha}}{N(\alpha)}.$$

Proof. The four matrix entries $x+s$, $x-s$, $r+t$, $r-t$ are independent linear forms in x,t,r,s since the characteristic is not two, so μ is a linear bijection. Both the product formula and matrix multiplication are bilinear and have $1 = (1,0)$ as identity, so it suffices to compare them on pure basis elements. The matrices satisfy $J^2 = -I$, $K^2 = D^2 = I$, and

$$KJ = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = D, \qquad KD = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = J, \qquad JD = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = K,$$

with $JK = -D$, $DK = -J$, $DJ = -K$. On the other side, for pure dihedrons the product formula gives $(0,v)\otimes(0,u) = (-\langle v,u\rangle\,, -v\times u)$; with the basis vectors e_1, e_2, e_3 of $\mathbb{F}^3$ one has

$\langle e_1, e_1\rangle = 1$, $\langle e_2, e_2\rangle = \langle e_3, e_3\rangle = -1$, and $e_1 \times e_2 = e_3$, $e_2 \times e_3 = e_1$, $e_3 \times e_1 = -e_2$, which yields exactly the table. So μ is multiplicative. The adjugate of $\mu(\alpha)$, obtained by swapping the diagonal entries and negating the others, is $\begin{pmatrix} x-s & t-r \\ -r-t & x+s \end{pmatrix} = \mu(\overline{\alpha})$. The remaining statements are the corresponding facts $M \operatorname{adj} M = (\det M) I$ and $\det(MM') = \det M \det M'$ for 2×2 matrices, together with associativity of matrix multiplication. ■

In coordinates, if $\alpha = (x_1, t_1, r_1, s_1)$ and $\beta = (x_2, t_2, r_2, s_2)$, then $\alpha \otimes \beta = (x, t, r, s)$ with

$$\begin{aligned} x &= x_1x_2 - t_1t_2 + r_1r_2 + s_1s_2, \\ t &= x_1t_2 + x_2t_1 + r_1s_2 - r_2s_1, \\ r &= x_1r_2 + x_2r_1 + t_1s_2 - s_1t_2, \\ s &= x_1s_2 + x_2s_1 + t_2r_1 - t_1r_2. \end{aligned}$$

Remark 75 *The dihedron algebra is the split quaternion algebra, also called the coquaternion algebra of Cockle [4], introduced six years after Hamilton's quaternions [7]. Our multiplication order is the one natural for row vectors acted on from the right. The opposite convention gives the usual Cockle product, and the opposite product $\otimes^{op}$ of Section 9. The Euclidean quaternion with coordinates (x, t, r, s) has, in the matrix model of [11], the matrix $\mu(x, t, \iota r, \iota s)$ with $\iota^2 = -1$: the two algebras are two $\mathbb{F}$-forms of $M_2(\mathbb{F}[\iota])$.*

11.1 Projective proportions

A **proportion** is a non-zero dihedron up to multiplication by a non-zero scalar:

$$(x : v) = (x : t : r : s).$$

It is non-null when

$$N(x, v) = x^2 + Q(v) = x^2 + t^2 - r^2 - s^2$$

is non-zero.

For a non-null proportion define

$$\rho(x : v) \equiv I + \frac{2(xA_v + A_v^2)}{x^2 + Q(v)}.$$

This is unchanged by rescaling the representative. On the chart $x \neq 0$,

$$\rho(x : v) = c\left(A_{v/x}\right).$$

At infinity, when $x = 0$ and v is non-null,

$$\rho(0 : v) = -I + \frac{2}{Q(v)} v^\dagger v,$$

the half turn about $[v]$. By the quadrance of an anchor product lemma, the polynomial identity $P = 0$ established in the proof of the composition theorem reads

$$c(A_v)\, c(A_u) = \rho(1 - \langle v, u\rangle : v + u - v \times u)$$

for all v, u with $Q(v), Q(u) \neq -1$, with no restriction on $\langle v, u\rangle$.

Theorem 76 (Half spread of a proportion) *For a non-null proportion $(x : v)$ with $N = x^2 + Q(v)$,*

$$p(\rho(x : v)) = \frac{Q(v)}{N}, \qquad 1 - p(\rho(x : v)) = \frac{x^2}{N},$$

and

$$\rho(x : -v) = \rho(x : v)^\dagger = \rho(x : v)^{-1}.$$

Proof. Since $\operatorname{tr} A_v = 0$ and $\operatorname{tr} A_v^2 = -2Q(v)$, $\operatorname{tr} \rho(x : v) = 3 - \frac{4Q(v)}{N}$, so $p = \frac{Q(v)}{N}$ and $1 - p = \frac{x^2}{N}$. Replacing v by $-v$ replaces A_v by $A_{-v} = -A_v = A_v^\dagger$ and leaves A_v^2 and N unchanged, which passes to the adjoint; and the adjoint of a rotation is its inverse. ∎

The product of proportions is the projectivization of the matrix product:

$$(x : v) \otimes (y : u) \equiv (xy - \langle v, u \rangle : xu + yv - v \times u).$$

Theorem 77 (Projective product) *The non-null proportions form a group under $\otimes$. The identity is $(1 : 0)$ and*

$$(x : v)^{-1} = (x : -v).$$

The product is associative and total on non-null proportions, and the quadrance class is multiplicative.

Proof. Under μ these are the invertible 2×2 matrices modulo non-zero scalars, and N is the determinant. ∎

Theorem 78 (Sandwich formula) *Let $\alpha = (x, v)$ represent a non-null proportion and identify $w \in \mathbb{F}^3$ with the pure dihedron $(0, w)$. Then*

$$w\,\rho(x : v) = \frac{\overline{\alpha} \otimes w \otimes \alpha}{N(\alpha)}.$$

The right side is again pure.

Proof. First $\overline{\alpha} \otimes w = (x, -v) \otimes (0, w) = (\langle v, w \rangle, xw + v \times w)$. Then

$$(\langle v, w \rangle, xw + v \times w) \otimes (x, v) = (x\langle v, w \rangle - \langle xw + v \times w, v \rangle, \langle v, w \rangle v + x(xw + v \times w) - (xw + v \times w) \times v).$$

The scalar part is $x\langle v, w \rangle - x\langle w, v \rangle - 0 = 0$. In the vector part, $-(xw) \times v = x(v \times w)$ and $-(v \times w) \times v = v \times (v \times w) = \langle v, w \rangle v - Q(v) w$ by the vector triple product identity, so the vector part is

$$\left(x^2 - Q(v)\right) w + 2x(v \times w) + 2\langle v, w \rangle v.$$

On the other side, by the cross product lemma and part 1 of the corollary of Section 3, $wA_v = v \times w$ and $wA_v^2 = \langle w, v \rangle v - Q(v) w$, so

$$N(\alpha)\; w\rho(x : v) = \left(x^2 + Q(v)\right) w + 2x(v \times w) + 2\langle w, v \rangle v - 2Q(v) w,$$

which is the same. ∎

Theorem 79 (Projective rotation theorem) *The map*

$$(x : v) \longmapsto \rho\,(x : v)$$

is a bijection from the non-null proportions of $P^3(\mathbb{F})$ *onto* $SO(1,2,\mathbb{F})$*, and*

$$\rho\,((x : v) \otimes (y : u)) = \rho\,(x : v)\,\rho\,(y : u)\,.$$

If $p(\rho) \neq 1$*, its inverse projective coordinate is*

$$\rho = \rho\,(1 + \operatorname{tr}\rho : w)\,, \qquad A_w = 2\rho^{as}.$$

For a half turn one may take $\rho = \rho\,(0 : v)$*, where* v *is any non-zero row of* $I + \rho$*.*

Proof. *Multiplicativity*: with representatives $\alpha = (x, v)$ and $\beta = (y, u)$, the sandwich formula applied twice, associativity, and $\overline{\alpha \otimes \beta} = \overline{\beta} \otimes \overline{\alpha}$ (the adjugate reverses products) give for any vector w

$$w\,\rho\,(\alpha)\,\rho\,(\beta) = \frac{\overline{\beta} \otimes (\overline{\alpha} \otimes w \otimes \alpha) \otimes \beta}{N(\alpha)\,N(\beta)} = \frac{\overline{\alpha \otimes \beta} \otimes w \otimes (\alpha \otimes \beta)}{N(\alpha \otimes \beta)} = w\,\rho\,(\alpha \otimes \beta)\,.$$

Surjectivity: a rotation with $p \neq 1$ is a cap (Section 7), hence $\rho\,(1 : w)$ with $A_w = A(\rho) = \frac{2}{1+\operatorname{tr}\rho}\rho^{as}$, which is the stated recipe after rescaling by $1 + \operatorname{tr}\rho$; a rotation with $p = 1$ is a half turn by the half turn lemma, which is $\rho\,(0 : v)$, and its axis vector spans the row space of $I + \rho_{[v]} = \frac{2}{Q(v)} v^{\dagger} v$. *Injectivity*: by the half spread of a proportion, caps ($x \neq 0$, $p \neq 1$) and half turns ($x = 0$, $p = 1$) do not mix. On $x \neq 0$, normalize $x = 1$: injectivity is part 5 of the theorem that caps are rotations. At infinity, distinct non-null lines give distinct half turns by the half turns theorem, and $(0 : v)\,, (0 : \lambda v)$ are the same proportion. ■

Corollary 80 (The Lorentz group is PGL_2) *For every field of characteristic not two,*

$$SO(1,2,\mathbb{F}) \cong PGL_2(\mathbb{F}) = GL_2(\mathbb{F})\,/\mathbb{F}^*,$$

where $\mathbb{F}^*$ *denotes the non-zero scalars. Over* $\mathbb{Q}$ *the rational Lorentz group is projective three space with the null quadric removed. Over a finite field* $\mathbb{F}_q$ *with* q *odd, the null proportions number* $(q+1)^2$ *and*

$$|SO(1,2,\mathbb{F}_q)| = q\left(q^2 - 1\right).$$

Proof. Under μ, non-null dihedrons are invertible matrices and proportions are matrices up to scalars. Over $\mathbb{F}_q$ the singular matrices number $q^4 - (q^2-1)(q^2-q) = q^3 + q^2 - q$ including zero, so the null proportions number $(q^3 + q^2 - q - 1)/(q-1) = (q+1)^2$, and $|PGL_2(\mathbb{F}_q)| = (q^2-1)(q^2-q)/(q-1) = q(q^2-1)$. ■

Over an ordered field the direction of time is detected by a single matrix entry. Call a rotation ρ **orthochronous** when the first coordinate of $(1,0,0)\,\rho$, namely the entry ρ_{11}, is positive; over $\mathbb{R}$ these are the transformations of the proper orthochronous Lorentz group $SO^+(1,2)$.

Lemma 81 (Time orientation) *For a non-null proportion $(x:t:r:s)$ with $N = x^2+t^2-r^2-s^2$,*

$$\rho\left(x:t:r:s\right)_{11} = \frac{x^2+t^2+r^2+s^2}{N}.$$

Consequently, over an ordered field, $\rho(x:v)$ is orthochronous precisely when $N > 0$, and the orthochronous rotations form a subgroup $SO^+(1,2,\mathbb{F})$.

Proof. The $(1,1)$ entry of A_v is 0. By part 1 of the corollary of Section 3, $A_v^2 = v^\dagger v - Q(v)\, I$, and $\left(v^\dagger v\right)_{11} = \left(Gv^T\right)_1 v_1 = t^2$, so $\left(A_v^2\right)_{11} = t^2 - Q(v) = r^2+s^2$. Hence

$$\rho_{11} = 1 + \frac{2\left(r^2+s^2\right)}{N} = \frac{x^2+t^2-r^2-s^2+2r^2+2s^2}{N} = \frac{x^2+t^2+r^2+s^2}{N}.$$

The numerator is positive for a non-zero proportion over an ordered field, so ρ_{11} has the sign of N. The sign of N is unchanged by rescaling, and is multiplicative by the projective product theorem, so the rotations with $N > 0$ form a subgroup. ■

For example, $\rho(3:0:0:1)_{11} = \frac{10}{8} = \frac{5}{4}$, while $\rho(2:1:2:3)_{11} = \frac{18}{-8} = -\frac{9}{4}$.

Over $\mathbb{R}$ the sign of $N = \det\mu$ is projectively well defined, and by the lemma the component $N > 0$ is $SO^+(1,2)$. A positive-determinant matrix can be normalized over $\mathbb{R}$ to determinant one, giving the usual double cover

$$SL_2(\mathbb{R}) \longrightarrow SO^+(1,2).$$

The component $N < 0$ is the non-orthochronous component of $SO(1,2,\mathbb{R})$.

11.2 The null quadric and spinors

The excluded projective points have $N = 0$, so their matrices have rank one.

Theorem 82 (Rulings of the null quadric) *A non-zero null dihedron has*

$$\mu(\alpha) = \kappa\varepsilon$$

for a non-zero column $\kappa \in \mathbb{F}^2$ and a non-zero row $\varepsilon \in \mathbb{F}^2$, each determined projectively. Holding $[\kappa]$ fixed gives one projective line on the null quadric; holding $[\varepsilon]$ fixed gives another. The two families are each parametrized by $P^1(\mathbb{F})$. Lines from the same family are disjoint and one line from each family meets in one point.

Proof. A non-zero singular 2×2 matrix has rank one, so its columns are proportional to a single column κ, and it is $\kappa\varepsilon$ with ε the row of coefficients; $(\lambda\kappa)(\lambda^{-1}\varepsilon)$ is the only ambiguity. For fixed κ, the matrices $\kappa\varepsilon$ with ε ranging over $\mathbb{F}^2$ form a two dimensional subspace, hence a projective line of proportions; likewise for fixed ε. Two matrices $\kappa\varepsilon, \kappa\varepsilon'$ with $[\varepsilon] \neq [\varepsilon']$ are not proportional, so lines of the same family are disjoint, while $\kappa\varepsilon$ is the unique common point of the lines of $[\kappa]$ and $[\varepsilon]$. The column lines and the row lines are each the points of $P^1(\mathbb{F})$. ■

The column and row lines are the two projective spinor families. Thus the spinors appear directly as the two rulings of the null quadric, as developed in [12]. This is the projective geometry behind the usual two-component spinor description [19].

The Euclidean companion has the same projective pattern but a different four-dimensional algebra. Over $\mathbb{Q}$ the Euclidean quaternion algebra is anisotropic, while the present split quaternion algebra is $M_2(\mathbb{Q})$. Over finite fields both split.

12 The obstruction over the rational numbers

Every rotation is represented by a non-null projective matrix. Its determinant is defined only up to a square, and this gives an arithmetic invariant. Write $(\mathbb{F}^*)^2$ for the non-zero squares, $\mathbb{F}^*/(\mathbb{F}^*)^2$ for the group of **square classes**, and $[c]$ for the class of a non-zero c.

Theorem 83 (Quadrance class) *The map*

$$\chi : SO(1,2,\mathbb{F}) \longrightarrow \mathbb{F}^*/(\mathbb{F}^*)^2$$

which sends a rotation to the square class of $N(x,v)$ for any projective representative is a well defined homomorphism. Moreover

$$\chi(\rho) = [1 - p(\rho)] \quad \text{if } p(\rho) \neq 1,$$

while for a half turn

$$\chi\left(\rho_{[v]}\right) = [Q(v)].$$

Proof. Rescaling a representative multiplies N by a square, and multiplication of proportions multiplies N. For a non-half-turn use a representative $(1 : v)$:

$$N = 1 + Q(v) = \frac{1}{1-p},$$

which has the same square class as $1 - p$. For a half turn use $(0 : v)$. ■

Corollary 84 (Spinor norm) *For non-null vectors v_1 and v_2,*

$$\chi(\sigma_{v_1}\sigma_{v_2}) = [Q(v_1)\,Q(v_2)].$$

Proof. Write $d \equiv \langle v_1, v_2\rangle$. If $d \neq 0$, the anchor of two reflections theorem gives the anchor vector $\frac{1}{d}v_1 \times v_2$, so by the Lagrange identity

$$\chi = \left[1 + \frac{Q(v_1 \times v_2)}{d^2}\right] = \left[\frac{Q(v_1)\,Q(v_2)}{d^2}\right] = [Q(v_1)\,Q(v_2)].$$

If $d = 0$, then v_1 and v_2 are independent, $v_1 \times v_2$ has quadrance $Q(v_1)\,Q(v_2) \neq 0$, and $\sigma_{v_1}\sigma_{v_2}$ fixes $v_1 \times v_2$ and negates v_1 and v_2, which span $(v_1 \times v_2)^\perp$. By the half turns theorem it is $\rho_{[v_1 \times v_2]}$, with class $[Q(v_1 \times v_2)] = [Q(v_1)\,Q(v_2)]$. ■

By the two reflections theorem this computes χ on every rotation, and identifies it with the classical **spinor norm** [17], [16].

Theorem 85 (Surjectivity over $\mathbb{Q}$) *The spinor norm*

$$\chi : SO(1,2,\mathbb{Q}) \longrightarrow \mathbb{Q}^*/(\mathbb{Q}^*)^2$$

is onto.

Proof. For any non-zero rational c,

$$N\left(\frac{c+1}{2},0,\frac{c-1}{2},0\right)=\left(\frac{c+1}{2}\right)^2-\left(\frac{c-1}{2}\right)^2=c.$$

Thus every square class occurs. ■

Corollary 86 (The rational normalization obstruction) *A rational Lorentz transformation has a rational determinant-one matrix representative precisely when its spinor norm is trivial. These transformations form the image of $PSL_2(\mathbb{Q})$ and a subgroup of infinite index in $SO(1,2,\mathbb{Q})$.*

Proof. A representative $\lambda\mu(\alpha)$ has determinant $\lambda^2 N(\alpha)$, which can equal 1 for rational λ precisely when $N(\alpha)$ is a square. The square classes of the primes, together with $[-1]$, are pairwise distinct, so the quotient by the kernel is infinite. ■

The same obstruction appears for normalized timelike split quaternions in [18]. The boost

$$\rho(3:0:0:1)$$

has $N=8$ and hence spinor norm [2]. It is a perfectly rational Lorentz transformation, but no rational rescaling of its 2×2 matrix has determinant one. Over $\mathbb{R}$ there is no problem: multiplication by $1/\sqrt{8}$ gives the usual element of $SL_2(\mathbb{R})$. The obstruction is arithmetic, not geometric.

12.1 Existence and causal character

Theorem 87 (Existence) *Let v be non-null and let $p\in\mathbb{F}$ with $p\neq 0,1$. A rotation with axis $[v]$ and half spread p exists precisely when*

$$Q(v)\,p(1-p)$$

is a non-zero square in $\mathbb{F}$. Equivalently, $Q(v)\,s$ is a square, where

$$s=4p(1-p)$$

is the rotation spread. When the condition holds there are exactly two such rotations, mutually inverse, with anchor vectors $\pm\lambda v$. For $p=1$ there is exactly one, the half turn $\rho_{[v]}$, by the half turn lemma.

Proof. A rotation with $p\neq 1$ is a cap $c(A_w)$, and $w\neq 0$ since $p\neq 0$. Its fixed vectors form exactly the line $[w]$: the eigenvalue 1 of the characteristic polynomial $(X-1)(X^2-2(1-2p)X+1)$ is simple, since the quadratic factor at $X=1$ equals $4p\neq 0$. So the axis condition forces $w=\lambda v$ with $\lambda\neq 0$, and the half twist gives

$$\lambda^2 Q(v)=Q(w)=\frac{p}{1-p}.$$

This is solvable for λ precisely when $\frac{p}{(1-p)Q(v)}$ is a square; multiplying by the non-zero square $((1-p)Q(v))^2$ identifies this class with that of $Q(v)\,p(1-p)$, and multiplying by 4 gives the

spread form. Conversely, when λ exists, $Q(\pm\lambda v) = \frac{p}{1-p} \neq -1$, so the caps $c(A_{\pm\lambda v})$ are defined, fix v, have half spread p by the half spread identities, and are mutually inverse. ■

For a non-half-turn $\rho \neq I$ we take as its axis the line of its anchor vector. Over an ordered field the causal type of this axis now gives the familiar physical classification.

Corollary 88 (Half spread and causal type) *Over an ordered field, let $\rho \neq I$ be an orthochronous rotation which is not a half turn. Then:*

1. *a timelike axis has $0 < p < 1$, and gives an ordinary spatial rotation;*
2. *a spacelike axis has $p < 0$, and gives a boost-type transformation;*
3. *a null axis has $p = 0$, and gives a parabolic or null rotation.*

A non-orthochronous rotation which is not a half turn has $p > 1$ and a spacelike axis.

Proof. Write $\rho = \rho(1 : v)$ with v the anchor vector, so $N = 1 + Q(v)$, $p = Q(v)/N$ and $1 - p = 1/N$. If ρ is orthochronous then $N > 0$ by the time orientation lemma, so $p < 1$ and p has the sign of $Q(v)$. If ρ is not orthochronous then $N < 0$, so $1 - p < 0$, that is $p > 1$, and $Q(v) = pN < 0$. ■

Thus the sign ranges have a direct physical meaning. The boost in the introduction has $p = -1/8$ and lies in $SO^+(1, 2)$. The spatial quarter turn has $p = 1/2$. By contrast

$$\rho(2 : 1 : 2 : 3)$$

has $N = -8$ and $p = 3/2$; it is not orthochronous and reverses time orientation, as its entry $\rho_{11} = -9/4$ shows.

12.2 Causal and arithmetic classification

There are two useful classifications, and they should not be confused.

Over $\mathbb{R}$, for an orthochronous rotation, the causal character of the fixed axis gives the physical classification:

$Q(v) > 0$	elliptic: spatial rotation
$Q(v) < 0$	hyperbolic: boost type
$Q(v) = 0$	parabolic: null rotation.

The non-orthochronous non-half-turns all have spacelike axes, by the previous corollary, but are not boosts: for instance $\rho(2 : 1 : 2 : 3)$ has eigenvalues 1 and $-2 \pm \sqrt{3}$.

Over a general field there is an additional arithmetic question: does the semisimple part split over the field?

Theorem 89 (Arithmetic splitting) *Let ρ have non-null axis $[v]$ and $p \neq 0, 1$. The quadratic factor of its characteristic polynomial,*

$$X^2 - 2(1 - 2p)X + 1,$$

has discriminant of square class

$$[-Q(v)].$$

*Hence the two non-trivial eigenvalues lie in $\mathbb{F}$ precisely when $-Q(v)$ is a square in $\mathbb{F}$. We call the transformation **split semisimple** in this case and **nonsplit semisimple** otherwise.*

Proof. The discriminant is

$$4\left(1-2p\right)^{2}-4=-16p\left(1-p\right).$$

By the existence theorem, $[p\left(1-p\right)]=[Q\left(v\right)]$, so the discriminant has class $[-Q\left(v\right)]$. ∎

For example, a spacelike axis with $-Q(v)$ a non-square is still hyperbolic over $\mathbb{R}$, but it is non-split over $\mathbb{Q}$. The axis $(1,2,3)$ has $Q=-12$, and 12 is not a rational square, so the orthochronous real transformations about this axis are boost-type although their non-trivial eigen-directions are not rational.

12.3 Ray proportions and spin

Over an ordered field a **ray proportion** is a non-zero (x,v) modulo positive rescaling. Every projective proportion contains exactly two rays. The dihedron product respects positive rescaling, so the non-null rays form a group and

$$\{\text{non-null rays}\}\longrightarrow SO\left(1,2,\mathbb{F}\right)$$

is two to one, with kernel the rays of 1 and -1.

Over $\mathbb{R}$ each ray with $N>0$ has a unique representative with $N=1$. Thus the positive-N ray group is naturally $SL_{2}\left(\mathbb{R}\right)$, and the map

$$SL_{2}\left(\mathbb{R}\right)\longrightarrow SO^{+}\left(1,2,\mathbb{R}\right)$$

is the usual spin double cover with kernel $\{\pm I\}$.

Over $\mathbb{Q}$ the situation is different. A rational ray can be normalized to determinant one over $\mathbb{Q}$ only when its determinant is a rational square. Thus the rational points of $Spin\left(1,2\right)\cong SL_{2}$ cover only $\ker\chi$, while rational rays give a two-to-one cover of the whole group $SO\left(1,2,\mathbb{Q}\right)$. The ray cover and the rational points of the spin group are therefore distinct arithmetic objects.

Rays also carry the orientation data of Section 8. On a non-null axis $[v]$ the two anchor vectors $\pm\lambda v$ give mutually inverse rotations, and the choice between them is a choice of ray on the anchor line. For timelike axes this is the classical sense of rotation; for spacelike and null axes, where no metrical determinant decides it, the ray is the invariant content of orientation.

This also clarifies the two rulings of the null quadric. The rank-one matrix $\kappa\varepsilon$ carries a column spinor and a row spinor. These are the projective spinor coordinates which survive on the null boundary.

13 Concluding remarks

13.1 What the projective viewpoint adds

The central formulas are classical. Rodrigues parameters, Cayley transforms, split quaternions and the identification of the real Lorentz group with a projective 2×2 matrix group all have long histories [2], [3], [4], [19]. The point here is their organization over a general field.

The half spread gives a field-valued measure of the amount of a transformation. The anchor is recovered directly from adjoint-antisymmetrization. Its composition law contains both the circular

tangent law and the hyperbolic half-rapidity law. Projectivization removes the half-turn singularity, the forbidden denominator and the excluded affine quadric at once. In the Minkowski case the resulting algebra is simply $M_2(\mathbb{F})$.

Over $\mathbb{Q}$ the projective formulation also reveals information which normalization hides. The determinant square class is the spinor norm and takes every value in $\mathbb{Q}^*/(\mathbb{Q}^*)^2$. Thus $PSL_2(\mathbb{Q})$ accounts for only the trivial spinor-norm class, while $PGL_2(\mathbb{Q})$ gives all rational Lorentz transformations.

The null quadric is equally natural in this language. It is the rank-one locus of M_2, and its two rulings are the column and row spinor families. Within the orthochronous group, the elliptic, hyperbolic and parabolic real types become the timelike, spacelike and null cases of the same anchor geometry, and the sign of the determinant N separates the orthochronous component from the other.

13.2 Computation

Every construction uses a small number of field operations. Composition is a 2×2 matrix product or the equivalent four component dihedron product. The axis and half spread are obtained from an adjoint-antisymmetric part and a trace. Equality of projective representatives is checked by cross multiplication. Exact arithmetic over $\mathbb{Q}$ is available throughout.

This does not imply that projective coordinates are always numerically superior to normalized floating-point coordinates. Their advantage here is structural: they stay in the ground field, avoid square-root normalization, and keep the exceptional loci visible.

13.3 Open problems

Problem 90 *Classify the operations*

$$v \otimes u = \frac{v + u + \gamma\, v \times u}{1 + \delta\, \langle v, u \rangle}$$

with $\gamma \neq 0$ which are associative wherever defined. Conjugating the anchor product and its opposite by the rescaling $v \mapsto \lambda v$ gives the associative cases $\gamma = \pm\lambda$, $\delta = -\lambda^2$. Is $\delta = -\gamma^2$ necessary apart from degenerate operations?

Problem 91 *Describe the finitely generated subgroups of* $\ker \chi$ *in* $SO(1, 2, \mathbb{Q})$*, and determine the spinor norms of finite-order elements and natural arithmetic subgroups.*

Problem 92 *Give an entirely projective description of the Thomas–Wigner rotation arising from two non-collinear boosts, including a closed formula for its half spread and its spinor norm in terms of the two boost anchors.*

Problem 93 *Replace $t^2 - r^2 - s^2$ by a general non-degenerate ternary quadratic form. The corresponding quaternion algebra should give the projective rotation group in the same way, with the present split algebra and the Euclidean quaternions of [11] as the two basic examples. The vector algebra needed for a general form is in [13].*

Problem 94 *Develop the integral theory using primitive representatives* $(x : t : r : s)$*, where composition becomes an arithmetic problem for integral* 2×2 *matrices and the split quadratic form* $x^2 + t^2 - r^2 - s^2$*.*

13.4 The projective viewpoint

A real physicist may describe a spatial rotation by an angle and a boost by a rapidity. The present theory keeps instead the algebraic data carried by the anchor and then passes to a projective 2×2 matrix. The familiar analytic parameters are recovered from these coordinates when the real field is chosen, but they are not needed for the definitions or the composition law.

This is exactly parallel to the Euclidean companion. There the projective algebra is the quaternion algebra; here it is the dihedron algebra M_2. In both cases normalization selects only part of the rational arithmetic, while projectivization gives the full rotation group.

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